A kinetic-impact deflection model

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to explain your model or result. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

impulse: net force acting over time; change in momentum

01 Warm-up: bring back an earlier idea

A 0.3 kg projectile at 50 m/s sticks to a stationary 2.7 kg cart.
Find common speed and the fraction of initial kinetic energy remaining.

Word help: initial = at the beginning of the interval; subscript i

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × lead time. Toy model: target mass is much larger; straight-line shift comparison. Ejecta can also transfer momentum in a real impact.

Worked reasoning: In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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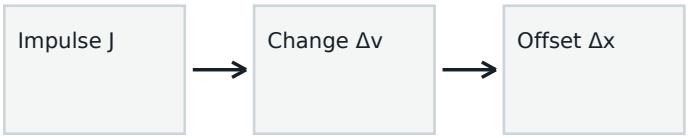
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

After a brief sideways push, a probe has a constant additional sideways velocity.
Compare the offset after a short drift and a long drift.
Use the model to connect impulse, velocity change and later position. Assume no further sideways force.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical 600 kg impactor moving at 6000 m/s sticks to a 5 × 10^9 kg target, initially at rest in the target frame. Ignore ejecta.
Estimate the target's velocity change and straight-line offset after 10^6 s.
Label the system before and after impact. These are model inputs, not mission measurements.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

Before

After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An asteroid's crossing window is days 16-20; Earth's is 18-22.
Find the overlap and explain why this is not proof of a collision.
Organize the timing evidence before drawing a conclusion.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Object	Arrival time / window	Comparison
A	___	___
B	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A model target needs a 10 km shift over 2×10^6 s. **Find** required Δv. If target mass 10^9 kg, estimate impulse. Organize the known quantities on the before-and-after model.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 1000 kg hypothetical target needs a transverse velocity change 0.02 m/s. **Estimate** required impulse. If delivered uniformly for 0.1 s, find average force and displacement from the velocity change after 10^5 s in a force-free simplified model. Organize the known quantities on the before-and-after model.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3